

JIANHAO MA

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RESEARCH INTERESTS

Continuous optimization; machine learning theory

EMPLOYMENT

Postdoctoral researcher

June 2025 - Now

Department of Statistics and Data Science, University of Pennsylvania

Supervisor: Prof. Yuxin Chen

EDUCATION

University of Michigan, Ann Arbor

January 2021 - May 2025

Ph.D. in Industrial and Operational Engineering

Advisor: Prof. Salar Fattahi

Tsinghua University

September 2016 - June 2020

B.E. in Industrial Engineering and B.S. in Mathematics

University of California, Berkeley

January 2019 - August 2019

Exchange student in the Department of Statistics

AWARDS

- ProQuest Distinguished Dissertation Awards, University of Michigan *2026*
- Rackham Predoctoral Fellowship, University of Michigan *2024-2025*
- INFORMS Junior Faculty Interest Group Paper Competition – Second Place (as a coauthor) *2023*
- Katta Murty Prize for Best Research Paper on Optimization, IOE Department *2023*
- NeurIPS Scholar Award *2022*

EXPERIENCE

FAIR Labs, Meta

May 2024 - August 2024

Research scientist intern, hosted by Dr. Lin Xiao

IIIS, Tsinghua University

August 2020 - June 2021

Visiting student, hosted by Prof. Yuhao Wang

AI Lab, ByteDance

April 2020 - July 2020

Machine learning engineer intern in deep reinforcement learning lab

PREPRINTS

(α - β : author names are sorted alphabetically)

1. Preconditioning Benefits of Spectral Orthogonalization in Muon [link]
Jianhao Ma, Yu Huang, Yuejie Chi, Yuxin Chen
2. Certifying Optimality in Nonconvex Robust PCA [link]
Pinxi Gong, Lexiao Lai, Jianhao Ma (α - β)

3. Quantization through Piecewise-Affine Regularization: Optimization and Statistical Guarantees [link]
Jianhao Ma, Lin Xiao
4. Implicit Regularization of Infinitesimally-perturbed Gradient Descent Toward Low-dimensional Solutions [link]
Jianhao Ma, Geyu Liang, Salar Fattahi
5. RANSAC Revisited: An Improved Algorithm for Robust Subspace Recovery under Adversarial and Noisy Corruptions [link]
Guixian Chen, Jianhao Ma, Salar Fattahi
6. LIBRA: Language Model Informed Bandit Recourse Algorithm for Personalized Treatment Planning [link]
Junyu Cao, Ruijiang Gao, Esmail Keyvanshokoh, Jianhao Ma (α - β)

PUBLICATIONS

(*: equal contribution)

1. Can Learning Be Explained By Local Optimality In Low-rank Matrix Recovery?
Jianhao Ma, Salar Fattahi
Mathematics of Operations Research, 2025 [link]
INFORMS Junior Faculty Interest Group Paper Competition – Second Place
2. PARQ: Piecewise-Affine Regularized Quantization
Lisa Jin, Jianhao Ma, Zechun Liu, Andrey Gromov, Aaron Defazio, Lin Xiao
International Conference on Machine Learning (ICML), 2025 [link]
3. Convergence of Gradient Descent with Small Initialization for Unregularized Matrix Completion
Jianhao Ma, Salar Fattahi
Conference on Learning Theory (COLT), 2024 [link]
4. Robust Sparse Mean Estimation via Incremental Learning
Jianhao Ma, Rui Ray Chen, Yinghui He, Salar Fattahi, Wei Hu
ICLR Workshop on Bridging the Gap Between Practice and Theory in Deep Learning, 2024 [link]
5. Global Convergence of Sub-gradient Method for Robust Matrix Recovery: Small Initialization, Noisy Measurements, and Over-parameterization
Jianhao Ma, Salar Fattahi
Journal of Machine Learning Research (JMLR), 2023 [link]
6. Behind the Scenes of Gradient Descent: A Trajectory Analysis via Basis Function Decomposition
Jianhao Ma, Lingjun Guo, Salar Fattahi
International Conference on Learning Representations (ICLR), 2023 [link]
7. Blessing of Nonconvexity in Deep Linear Models: Depth Flattens the Optimization Landscape Around the True Solution
Jianhao Ma, Salar Fattahi
Advances in Neural Information Processing Systems (NeurIPS), 2022 (**Spotlight**) [link]
Katta Murty Prize for Best Research Paper on Optimization
8. Towards Understanding Generalization via Decomposing Excess Risk Dynamics
Jiaye Teng*, Jianhao Ma*, Yang Yuan
International Conference on Learning Representations (ICLR), 2022 [link]
9. Sign-RIP: A Robust Restricted Isometry Property for Low-rank Matrix Recovery
Jianhao Ma, Salar Fattahi
NeurIPS Workshop on Optimization for Machine Learning, 2021 [link]

INVITED TALK/PRESENTATION

1. **Annual Conference on Learning Theory**, Edmonton, July 2024
“Convergence of Gradient Descent with Small Initialization for Unregularized Matrix Completion”
2. **Peking University**, Center for Machine Learning Research, Beijing, April 2024
“Robust Matrix Recovery through Nonconvex Optimization: Challenges and Promises”
3. **The Chinese University of Hong Kong**, SEEM Seminar Series, Hong Kong, April 2024
“Robust Matrix Recovery through Nonconvex Optimization: Challenges and Promises”
4. **INFORMS Optimization Society Conference**, Houston, TX, March 2024
“Convergence of Gradient Descent with Small Initialization for Unregularized Matrix Completion”
5. **INFORMS Annual Meeting**, Phoenix, AZ, October 2023
“Behind the Scenes of Gradient Descent: A Trajectory Analysis via Basis Function Decomposition”
6. **ICSA Applied Statistics Symposium**, Ann Arbor, MI, June 2023
“Robust Sparse Mean Estimation via Incremental Learning”
7. **INFORMS Annual Meeting**, Indianapolis, IN, October 2022
“Blessing of Nonconvexity in Deep Linear Models: Depth Flattens the Optimization Landscape Around the True Solution”
8. **INFORMS Optimization Society Conference**, Greenville, SC, March 2022
“Global Convergence of Sub-gradient Method for Robust Matrix Recovery: Small Initialization, Noisy Measurements, and Over-parameterization”
9. **INFORMS Annual Meeting**, Anaheim, CA, October 2021
“Sign-RIP: A Robust Restricted Isometry Property for Low-rank Matrix Recovery”
10. **MOPTA Conference**, Bethlehem, PA, August 2021
“Sign-RIP: A Robust Restricted Isometry Property for Low-rank Matrix Recovery”

ACTIVITIES/ACADEMIC SERVICE

Organizer

Session chair: INFORMS Annual Meeting 2021, 2022, 2024

Reviewer

Journal: IEEE Transactions on Information Theory, IEEE Transactions on Signal Processing, SIAM Journal on Optimization, SIAM Review, JMLR.

Conference: ICML, NeurIPS, ICLR, AISTATS, NeurIPS Workshop on Optimization for Machine Learning, ICLR Workshop on Bridging the Gap Between Practice and Theory in Deep Learning.